

Multimodal Rumor Detection on Weibo

Enhancing Trust on Social Media Platforms using Deep Learning and Cross-Modal Fusion Techniques

by

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Agenda

01. Problem & Business Impact

Understanding challenges in rumor detection and platform implications for social media trust and integrity.

02. Technical Approach

Multimodal fusion strategies using BERT and ResNet18 architectures with corrected implementations

03. Results & Analysis

Performance metrics comparison, best model evaluation, and comprehensive analysis of findings.

04. Recommendations & References

Strategic deployment recommendations, future directions, and references for rumor detection.

Problem Statement & Business Impact

The Challenge

Rumors on social media platforms spread using misleading text and reused images simultaneously, making detection with single modality (text-only or image-only) insufficient



Business Impact

Platform reputation damage, user trust erosion, regulatory compliance risks, and brand credibility threats from undetected false information spreading rapidly.



Key Objectives

Develop multimodal deep learning model fusing text and image data for accurate rumor detection, enabling early intervention and preventing misinformation spread.



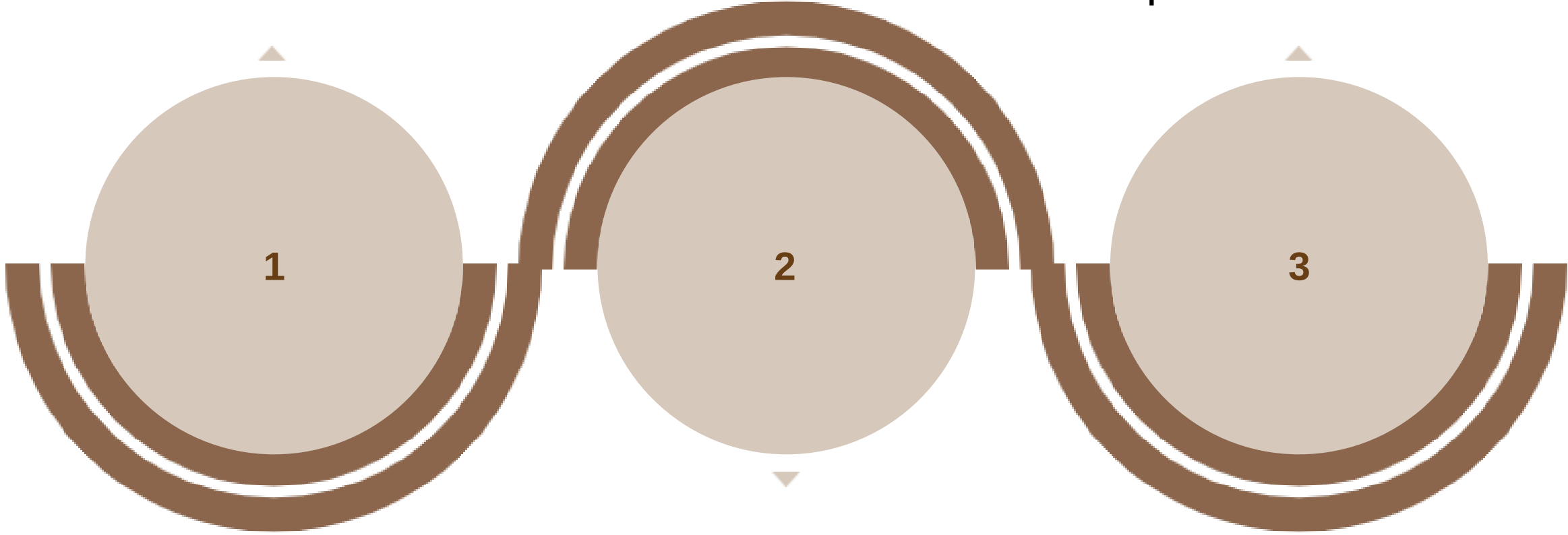
Complete Workflow Flowchart

Input Data Collection

Weibo posts with associated text and images labeled as Rumor or Non-Rumor categories for supervised learning.

Model Training & Evaluation

Train baseline and fusion models with comprehensive evaluation metrics including accuracy, AUC, F1, and visualization of performance results.



Data Preprocessing

BERT tokenization (max 128 tokens) for text and ResNet18 image transforms (224×224 pixels) with normalization.

Detailed Workflow Pipeline Architecture

5 Steps

Pipeline Stages

Step 1-2: Data & Preprocessing

JSON/CSV parsing, text cleaning, BERT tokenization, image resizing (224×224), normalization with ImageNet mean/std, and stratified 80/20 split.

- Text: max 128 tokens with padding and attention masks

5 Models

Baseline + Fusion

Step 3-4: Feature & Fusion

BERT extracts contextual text embeddings (768-d) and ResNet18 extracts visual embeddings (512-d) with optimal multimodal representation.

- Three fusion strategies: Late, Early, and Attention-based

6 Metrics

Evaluation

Step 5: Classification Head

Dense layers with ReLU activation, dropout regularization (0.3), and softmax for rumor/non-rumor classification.

Model Architectures: Comprehensive Overview

Baseline Models

Single-modality baseline approaches for performance comparison and understanding individual modality contributions to multimodal fusion.

- Text-only BERT: Hyper Parameter : LR=2e-5, Dropout=0.2, Epochs=3, Optimizer: AdamW
- Image-only ResNet18: Hyper Parameter : LR=1e-4, Dropout=0.3, Epochs=5, Optimizer :Adam
- Single modality insufficient for optimal rumor detection

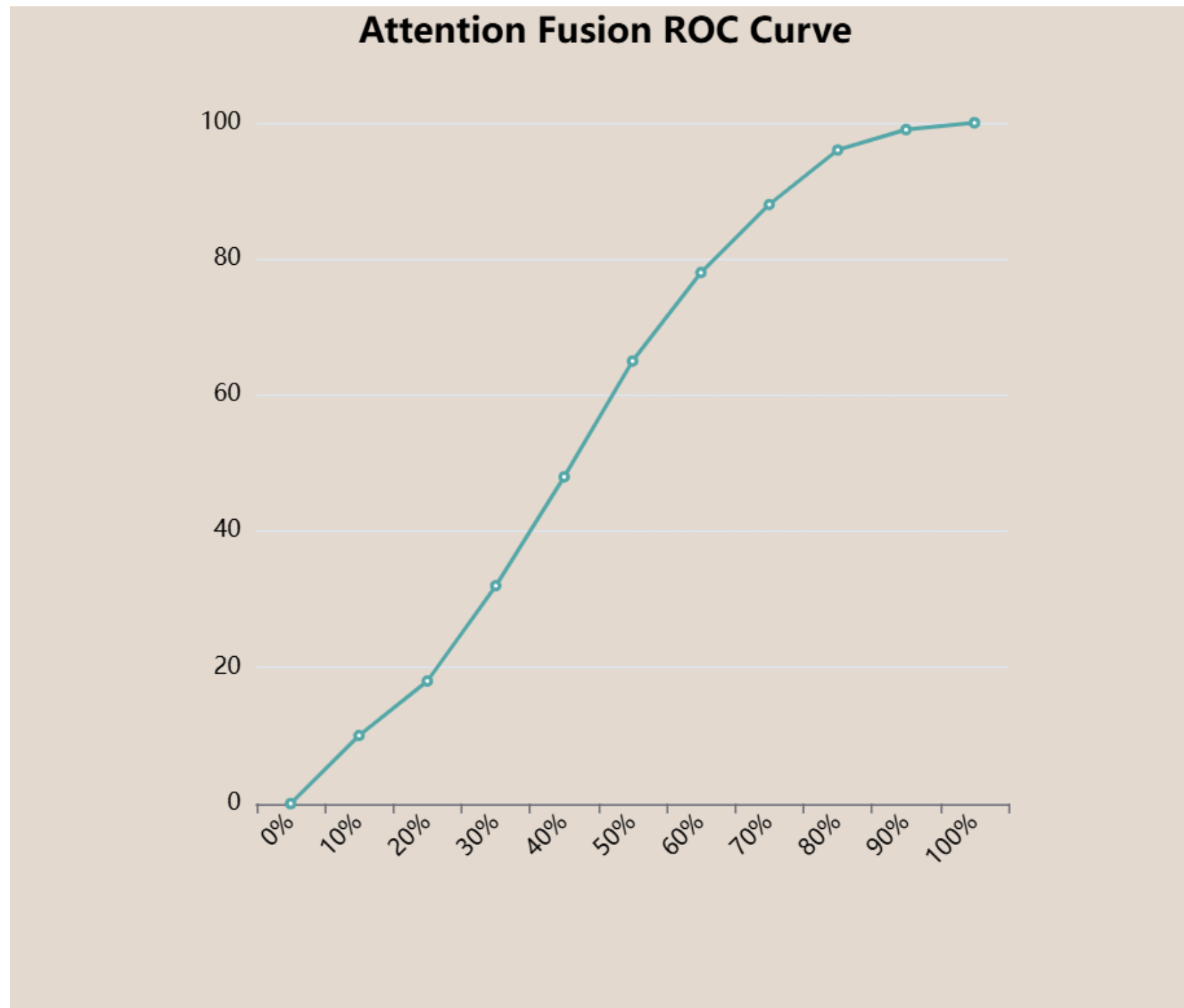
Fusion Models

- Late Fusion: Text (768-d) and Image (512-d) embeddings directly concatenated into 1280-d representation. Hyperparameters: LR=1e-4, Dropout=0.3, Epochs=5, |Optimizer :AdamW
- Early Fusion: Both embeddings projected to 512-d shared space before concatenation creating 1024-d unified representation. Hyperparameters: LR=1e-4, Dropout=0.1, 1 Encoder layer (8 heads), Epochs=5, Optimizer :AdamW
- Attention Fusion: Attended embedding (768-d) feeds into Dense 768→256 with ReLU activation Hyper parameter: LR=1e-4, Dropout=0.1, 8 Attention heads, Epochs=5, Optimizer :AdamW

Complete Performance Analysis Table

Model Configuration	Accuracy (%)	Macro-F1 (%)	Weighted-F1 (%)	AUC (%)	Average Precision (%)
Text-only BERT	97.40	94.25	97.51	97.71	99.73
Image-only ResNet18	88.31	62.19	86.27	54.17	89.45
Late Fusion	90.91	70.59	89.32	97.06	99.64
Early Fusion	90.91	70.59	89.32	96.08	99.48
🏆 Attention Fusion (BEST)	97.40	94.25	97.51	99.02	99.88

🏆 Attention Fusion: Overall Performance metrics



- 🏆 AUC Score: 99.02%
- 📊 Accuracy: 97.40%
- 🎯 Macro-F1: 94.25%

Attention Fusion: Confusion Matrix & Analysis

True \ Predicted	Predicted Non Rumor	Predicted Rumor	Class Total
Actual Non Rumor	TN=9	FP=0	9
Actual Rumor	FN=2	TP=66	68
Prediction Total	11	66	77

Classification Performance

- True Negatives (Non-Rumor): 9/9 (100% recall for non-rumor class)
- True Positives (Rumor): 66/68 (97.06% recall for rumor class)
- False Positives: 0 (0% false alarm rate for rumor detection)
- False Negatives: 2 (2.94% missed rumor detection rate)

Critical Findings

Key Performance Insights

Both Text-only BERT and Attention Fusion achieved very high performance with macro F1 ≈ 0.94 , demonstrating strong predictive capability.

- Attention Fusion Outperforms as Superior across all six metrics
- Early Fusion struggled due to noisy joint representation and overfitting on limited data.
- Late Fusion underperformed as text and image features didn't complement strongly.

Visual Features Complement Text

- Improved model robustness
- Enhanced generalizability

Class Imbalance Solutions

Oversampling and threshold tuning effectively improved fairness across classes, enhancing model performance

Future Research Directions

- Exhaustive hyper parameter tuning
- Build upon Attention Fusion success by exploring multi-head attention, self-attention, and hybrid mechanisms for further performance enhancement.
- Leverage Attention Fusion architecture with advanced training techniques and expanded datasets for improved generalization and robustness.
- Transfer learning from related social media platforms
- Explore CLIP and VisualBERT architectures
- Deploy Attention Fusion as primary model

References

Code and Data Source Link

<https://github.com/nupurmohanty16-debug/Multi-Modal-Rumor-Detection>

Dataset Source

- <https://drive.google.com/file/d/14VQ7EWPIFeGzxp3XC2DeEHIBEisDINn/view?usp=sharing>
- J. Devlin, M.-W. Chang, K. Lee and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *Proc. NAACL-HLT 2019*, pp. 4171–4186.
- M. Mosbach, M. Andriushchenko and D. Klakow, "On the Stability of Fine-Tuning BERT: Misconceptions, Explanations, and Strong Baselines," *Proc. ICLR 2021*.
- A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola, *Dive into Deep Learning*. Cambridge University Press, 2021.
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- J. M. Pérez-Rúa, V. Vielzeuf, S. Pateux, M. Baccouche and F. Jurie, "MFAS: Multimodal Fusion Architecture Search," *Proc. CVPR 2019*, pp. 6966–6975.

A close-up photograph of a hand holding a blue and silver pen, poised to write on a spiral-bound notebook. The background is softly blurred, showing other papers and a cup. The text is overlaid on the center of the image.

Thank You for Your Attention

Questions, Feedback, and Discussion Welcome